

AI-Based Detection of Apples for Autonomous Harvesting Systems

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Abstract

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Zusammenfassung

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List of Abbreviations

AI Artificial Intelligence

AP Average Precision

CNN Convolutional Neural Network

COCO Common Objects in Context

CV Computer Vision

DETR DEtection TRansformer

DL Deep Learning

FN False Negatives

FP False Positives

IoU Intersection over Union

mAP mean Average Precision

ML Machine Learning

NMS Non-Maximum Suppression

RPN Region Proposal Network

SSD Single Shot MultiBox Detector

TP True Positives

YOLO You Only Look Once

1 Introduction

This chapter introduces the motivation and context of the thesis. First, the general background and relevance of AI-based autonomous harvesting systems are presented. Subsequently, the specific challenges and research problems addressed in this work are described. Finally, the objectives and scope of the thesis are defined.

1.1 Background and Context

Agricultural production is increasingly influenced by automation and the application of artificial intelligence in order to improve efficiency, reduce manual labor requirements, and enable more sustainable production processes. In particular, fruit harvesting represents a highly complex task for robotic systems due to the unstructured and dynamic nature of orchard environments.

Within this context, autonomous harvesting systems aim to combine robotics, sensor technology, and computer vision to enable the automatic detection, localization, and harvesting of fruits. A central requirement for such systems is the ability to accurately identify and spatially localize fruits within camera-based perception systems, as this information is essential for subsequent robotic manipulation.

This thesis is conducted within the research project ANNA (Autonomous Harvesting and Thinning), a collaborative initiative involving industrial and academic partners. The project focuses on the development of an autonomous apple harvesting system that integrates robotic platforms with AI-based perception modules.

The work is carried out in cooperation between the DHBW Ravensburg (Campus Friedrichshafen) and the IWT (Institute for Continuing Education, Knowledge and Technology Transfer), which acts as a research and transfer institution connecting academic research with industrial application domains.

1.2 Problem Statement

Although deep learning-based object detection systems have shown strong performance in controlled environments, reliable fruit detection in real orchard scenarios remains a challenging task. Apples can appear under varying lighting conditions, may be partially occluded by leaves or branches, and can differ in color, size, and visibility depending on their maturity and position within the tree canopy. These factors complicate the detection and localization process and reduce the robustness of AI-based harvesting systems.

In addition to environmental challenges, the creation of large-scale real-world datasets represents a significant limitation for the development of reliable detection models. Acquiring image data in orchards requires considerable effort and is restricted by seasonal availability and environmental conditions. Furthermore, the annotation process is labor-intensive and costly, particularly when large amounts of training data are required.

Synthetic data generated from simulation environments offers a potential solution to these limitations. Simulated datasets can be generated efficiently and allow controlled variation of environmental parameters such as lighting conditions, camera perspectives, and object placement. However, models trained exclusively on synthetic data often suffer from limited generalization capabilities when applied to real-world environments due to the domain gap between simulated and real images.

Consequently, an important research question is to what extent synthetic data can contribute to the development of reliable apple detection systems for autonomous harvesting applications. In particular, it remains unclear whether hybrid training approaches combining real and synthetic datasets can improve detection performance and generalization capabilities compared to models trained solely on real or synthetic data.

1.3 Objectives of the Thesis

The objective of this thesis is to develop and evaluate an AI-based pipeline for the detection and localization of apples in orchard environments as part of an autonomous harvesting system. The work focuses on the generation and preparation of training datasets as well as the investigation of different training strategies for deep learning-based object detection.

To achieve this objective, image data is collected both in a real apple orchard and from an existing simulation environment of apple trees. Based on these datasets, a labeling process is performed to create annotated training data for model development. Subsequently, different AI models are trained and evaluated using three different dataset configurations: real-world data only, synthetic data only, and a hybrid combination of both data sources.

The developed models are analyzed and compared with regard to their detection and localization performance. Particular attention is given to the influence of synthetic training data and its potential to improve the robustness and generalization capabilities of AI-based apple detection systems.

This thesis focuses exclusively on the computer vision and AI-related components of the harvesting system. Mechanical aspects of the robotic platform and the physical harvesting process are not part of this work.

2 Fundamentals

This chapter provides the theoretical foundations that are required for this work. It introduces the basic concepts of deep learning and convolutional neural networks and the principles of object detection. Additionally it presents relevant model architectures, and defines the evaluation metrics used to assess model performance.

2.1 Deep Learning and CNNs

Artificial Intelligence (AI) includes a wide range of systems, that are built to automate routine work and understand complex data, such as images. [2, p. 1] Over time, AI evolved from rule-based systems and manually encoded knowledge toward Machine Learning (ML), in which models learn patterns directly from raw data instead of relying on explicit instructions. [2, p. 2] Deep Learning (DL) is a specialized subset of ML and extends this idea by modeling the real world as a hierarchical structure of increasingly abstract concepts. [2, p. 8] These models are most commonly trained through supervised learning. Using supervised learning, the algorithm receives input features (x) together with their corresponding target labels (y) and learns a function that maps one to the other. [2, p. 105] These tasks are typically divided into classification, which assigns inputs to discrete categories, and regression, which predicts continuous numerical values. [2, p. 100-101] Model training involves minimizing a loss function using an optimization algorithm, most commonly stochastic gradient descent, with the objective of achieving strong generalization to previously unseen data. [2, p. 15, 110] Computer Vision (CV) focuses on enabling machines to understand and interpret visual data such as images. For such tasks, CNNs are the dominant architecture, as they are specifically designed to process data with a grid-like topology, such as digital images. [2, p. 330] Unlike fully connected networks, CNNs use local connectivity and shared parameters, meaning each neuron processes only a small region of the input while the same filters are applied across the entire image. This design greatly reduces

the number of parameters and ensures that features can be recognized regardless of their position. [3, p. 13-14] CNNs are powerful because they automatically learn hierarchical representations of data. Early layers detect simple patterns like edges and colors, while deeper layers combine these into more complex features such as object parts and full objects. [3, p. 2] Pooling layers further improve efficiency by downsampling the data, making the model more robust to small shifts and focusing on whether a feature is present rather than exactly where it appears. [3, p. 17]

2.2 Object Detection

Object detection is a fundamental task in computer vision that involves identifying and localizing specific instances of objects within digital images. Unlike simple image classification, which assigns a single label to an entire image, object detection aims to answer the question "what objects are where?" by providing both a class label and a spatial definition, typically in the form of a bounding box. This process serves as the basis for complex tasks such as instance segmentation and object tracking. [4, p. 1] While early detection methods relied on handcrafted features and sophisticated speed-up skills, the field has been revolutionized by deep learning and CNNs. Modern deep learning-based detectors are traditionally divided into two categories: two-stage and one-stage detectors. [4, p. 3] More recent approaches additionally include transformer-based architectures, which formulate detection as an end-to-end set prediction problem. Two-stage detectors like Faster R-CNN work in two steps: first, a Region Proposal Network (RPN) identifies potential object locations, and then a second stage refines these areas and classifies the objects. [5, p. 1-3] Conversely, one-stage detectors such as You Only Look Once (YOLO) treat detection as a single regression problem, predicting bounding boxes and class probabilities simultaneously for the entire image in a single pass. [4, p. 3] The choice between these paradigms often involves a trade-off between accuracy and speed. [4, p. 1] Two-stage models typically offer higher localization accuracy, whereas one-stage models are often preferred for real-time applications, especially on resource-constrained hardware. [4, p. 3-4]

2.3 Relevant Models

Building on the fundamental detection paradigms and the speed-accuracy trade-offs described in section 2.2, modern computer vision research has been defined by several key model families that established current standards for performance and flexibility. Among two-stage detectors, Faster R-CNN was highly influential for introducing the RPN, which allows the model to learn to propose object candidates within the same convolutional network used for classification. This innovation unified the detection pipeline and significantly reduced the computational bottleneck associated with previous external proposal methods. [4, p. 3] Mask R-CNN subsequently extended this framework by adding a parallel branch for predicting pixel-level masks, effectively evolving bounding-box detection into instance segmentation. [6, p. 1] By incorporating the RoIAlign layer to maintain exact spatial locations, Mask R-CNN improved localization accuracy, making it a robust baseline for tasks requiring precise object boundaries. [6, pp. 1, 6] In the domain of one-stage detectors, YOLO family prioritized real-time inference by framing object detection as a single regression problem. By predicting bounding boxes and class probabilities simultaneously from the entire image in a single pass, YOLO enabled high-speed applications that were previously unattainable for two-stage architectures. [4, p. 4] While early one-stage models faced challenges with localization and small objects, the Single Shot MultiBox Detector (SSD) improved multi-scale detection by introducing multi-resolution techniques. Unlike previous models that performed detection only on the final network layer, SSD utilizes multiple feature maps of different resolutions to detect objects of various sizes, a design choice that significantly boosted the accuracy of one-stage detectors. [4, pp. 4, 7] A more recent paradigm shift in the field is represented by DETection TRansformer (DETR), which applies the Transformer architecture to object detection. DETR distinguishes itself by viewing detection as a direct set prediction problem, which effectively removes the need for hand-designed components commonly used in traditional detection pipelines. These include spatial anchors, which are predefined reference boxes for estimating object locations and scales, and Non-Maximum Suppression (NMS) post-processing,

which suppresses duplicate overlapping detections. [4, p. 4] [7, p. 1] By taking advantage of self-attention mechanisms to reason about the global image context and the relations between objects, DETR provides a streamlined, end-to-end framework that simplifies the detection pipeline while achieving performance comparable to highly optimized convolutional baselines. Self-attention enables the model to relate information from different regions of an image simultaneously, allowing it to better capture contextual dependencies and interactions between objects. [7, p. 2] These model families represent the primary architectural foundations upon which specialized applications, such as the detection of agricultural products in complex environments, are currently built.

2.4 Evaluation Metrics

Object detection requires a dual assessment of classification accuracy and localization precision to answer the fundamental questions of what objects are present and where they are located. The primary measure for localization quality is Intersection over Union (IoU). IoU calculates the ratio of the overlap area between the predicted bounding box and the ground-truth box to the area of their union. This metric serves as a localization quality measure, where a value of one indicates a perfect match. In standard evaluation protocols, an IoU threshold is set to determine whether a prediction is a correct detection or a miss. This threshold is most often set to 0.5.

Once detections are categorized based on IoU thresholds, performance is quantified using Precision and Recall. Precision measures the fraction of detections reported by the model that are correct (True Positives (TP)), whereas Recall measures the fraction of all actual ground-truth objects that were successfully identified. A model with high precision makes few false alarms (False Positives (FP)), while high recall indicates it misses few real objects (False Negatives (FN)). Because these metrics often involve a trade-off, where increasing one may decrease the other, the F1-Score is frequently used as a balanced metric, representing the harmonic mean of precision and recall.

To provide a comprehensive summary of a model's performance across all classification thresholds, the Average Precision (AP) is calculated as the area under the precision-recall curve. For multi-category detection tasks, the mean Average Precision (mAP) is the standard evaluation metric, representing the average AP across all object classes. Historically, benchmarks relied on a fixed IoU threshold (denoted as mAP@0.5), which identifies a detection as correct if it overlaps the ground truth by at least 50

Modern object detection systems increasingly rely on the more stringent Common Objects in Context (COCO) evaluation metrics to compare model performance. The COCO standard averages mAP across a range of ten IoU thresholds (from 0.50 to 0.95 in 0.05 increments), a method that specifically encourages and rewards superior localization accuracy. Furthermore, the COCO standard provides specialized metrics for different object sizes, allowing researchers to assess how effectively a model handles scale variations, which is critical for practical applications like apple detection in varying orchard environments.

3 State of the Art

Zusammenfassender Text

3.1 Evolution of AI in Agriculture

The application of artificial intelligence in agriculture has undergone a fundamental transformation over the past decades. This transformation is embedded in the broader evolution of agricultural production itself, which has progressed through several stages. Agriculture 1.0 relied on human and animal labor, while Agriculture 2.0 introduced mechanization and chemical inputs. Agriculture 3.0 brought digital tools such as GPS and IoT for precision farming. Today, Agriculture 4.0 integrates AI-driven analytics, robotics, and advanced sensing to create efficient, sustainable, and resilient food production systems [8, p. 1]. Within this context, early AI systems for agriculture relied on rule-based approaches and handcrafted features, whereas modern solutions leverage deep learning and large-scale pre-trained models. This evolution mirrors the general progress in computer vision and machine learning, but it has been shaped by the unique constraints of agricultural environments: high variability in lighting and weather, occlusions by foliage, overlapping fruits, and the need for real-time processing on resource-limited hardware such as agricultural robots [9, pp. 2-3].

3.1.1 From Traditional ML to DL

Before the widespread adoption of deep learning, fruit detection and recognition were predominantly based on traditional image processing and machine learning techniques. Researchers used manually designed features, such as color histograms in RGB, HSV, or Lab color spaces, texture descriptors, and shape features, which were then fed into classifiers like support vector machines, k-nearest neighbors, or decision trees [9, p. 2-3]. For example, early work on citrus harvesting employed chromaticity and bright-

ness information to separate fruits from leaves, achieving only moderate accuracy due to sensitivity to lighting changes [10, p. 12].

These traditional approaches suffered from several inherent limitations. First, the feature extraction process was problem-specific and required substantial engineering effort. Second, the algorithms were brittle under natural scene variations, such as shadows, backlight, or partial occlusions. Third, sliding-window-based detection was computationally expensive and produced many redundant candidate regions [10, p. 2]. Consequently, classical methods struggled to meet the real-time requirements of autonomous harvesting robots.

The breakthrough came with the introduction of deep convolutional neural networks. Instead of relying on hand-crafted features, CNNs learn hierarchical representations directly from raw image data. Early layers detect simple patterns like edges and colors, while deeper layers combine them into object parts and complete objects, as already deeper explained in Section 2. This capability proved especially valuable in agriculture, where fruit appearance varies considerably across cultivars, growth stages, and environmental conditions. As a result, deep learning-based detectors rapidly outperformed traditional methods in tasks such as fruit detection, disease identification, and yield estimation [9, pp. 2-3].

3.1.2 Vision Transformers and Foundation Models

For many years, convolutional neural networks, particularly the YOLO and Faster R-CNN families, dominated fruit detection research due to their favorable trade-off between accuracy and inference speed [9, p. 3-5]. More recently, two new trends have emerged: vision transformers and foundation models.

Vision transformers apply the self-attention mechanism, originally developed for natural language processing, to image recognition tasks. By treating an image as a sequence of patches, ViTs can capture long-range dependencies and global context more effectively than CNNs, which have a limited receptive field [11, p. 4]. In agricultural applications, ViTs have been explored for plant disease classification, crop type mapping, and

pest recognition, often outperforming traditional CNN architectures, especially when large training datasets are available [8, p. 12-13].

Foundation models represent a paradigm shift: they are pre-trained on massive, diverse datasets and can be adapted to many downstream tasks with little or no task-specific training. In agriculture, models such as SAM (Segment Anything Model) and CLIP have demonstrated impressive zero-shot segmentation and open-vocabulary detection capabilities [12, p. 3-4]. However, their enormous computational requirements and limited domain specificity pose challenges for practical deployment on farm machinery or embedded devices. Recent work addresses these issues by distilling knowledge from foundation models into smaller, faster student models, enabling real-time inference while retaining much of the generalization ability [12, p. 12-14].

Despite these advances, the adoption of large vision and language models in agriculture is still in its early stages. Bibliometric analyses show that agriculture-related keywords appear infrequently in the foundation model literature, indicating a significant research gap [11, p. 8]. This gap is particularly pronounced for object detection tasks that require high recall under occlusion and that must operate on edge hardware—exactly the conditions encountered by a robotic apple harvester.

3.2 Fruit Detection Research

Reliable fruit detection is a prerequisite for any automated harvesting system. The vision system must not only recognize the presence of a fruit but also accurately localize it in three-dimensional space to guide the robotic manipulator. However, orchards are unstructured environments that present a range of challenges rarely encountered in industrial or laboratory settings. This section reviews the principal difficulties, the sensor technologies developed to address them, and the deep learning approaches that have become the state of the art.

3.2.1 Environmental Challenges

The natural orchard environment imposes several interconnected challenges on fruit detection algorithms. Variable illumination is among the most persistent issues. Sunlight changes in intensity and direction throughout the day, producing strong shadows, backlighting, and specular reflections on fruit surfaces. Under such conditions, color-based segmentation methods that work reliably in controlled lighting often fail [13, p. 2].

Occlusion is a second major difficulty. Fruits are frequently hidden behind branches or leaves, or they may overlap with one another in dense clusters. In such cases, the visible portion of a target fruit may be reduced to a small fraction of its full extent, making detection and accurate bounding box placement difficult [14, p. 2]. A related problem is the presence of distractor objects that share visual properties with the target fruit, such as similarly colored leaves or stems, which can trigger false positives.

A third challenge is scale variation. Apples and other fruits appear at different sizes depending on their distance from the camera, their growth stage, and the camera's optical parameters. A detection model must therefore handle a wide range of scales, from close-up views of a single fruit to wide-angle scenes containing many small targets [9, p. 3]. Finally, motion blur caused by the movement of the robot platform or by wind-induced oscillation of branches can degrade image quality and further reduce detection reliability [13, p. 2+16]. These challenges are not independent, instead they often occur simultaneously, demanding robust and generalizable detection methods.

3.2.2 Sensor Technologies

The choice of sensor directly influences which visual information is available to the detection pipeline. Early fruit detection systems relied exclusively on standard RGB cameras, which provide color images at relatively low cost. RGB sensors capture texture and color features that are highly informative for ripe fruits, which often exhibit distinctive hues. However, RGB images lack direct depth information, making it impossible to

estimate fruit position in three-dimensional space without additional processing, such as structure from motion or stereo matching [13, pp. 4-5].

To obtain depth information, many harvesting robots employ stereo vision systems using two or more calibrated cameras. By triangulating corresponding points in the left and right images, stereo matching algorithms compute disparity maps, which can be converted to depth maps. While stereo vision is well understood and does not require active illumination, it is computationally expensive and can fail in texture-poor regions or under uniform lighting [9, pp. 10-11]. An alternative is the use of RGB-D cameras which combine an RGB sensor with a depth sensor based on structured light or time-of-flight. These devices provide registered color and depth images at video frame rates, greatly simplifying 3D fruit localization. Their main limitations are limited range, typically up to 3–5 meters, sensitivity to strong ambient infrared light, e.g. direct sunlight, and higher cost compared to simple RGB cameras [13, p. 4, p. 11-12].

For applications requiring long-range or highly accurate depth measurements, LiDAR sensors have been explored. LiDAR provides dense point clouds that can be used to reconstruct fruit shapes and estimate their positions even under challenging lighting conditions. However, LiDAR systems are expensive and their data processing is computationally intensive, which has limited their adoption in cost-sensitive agricultural robots [9, p. 13-14]. In practice, many recent fruit detection systems opt for a hybrid approach: an RGB camera for appearance-based detection and an RGB-D camera or stereo pair for depth estimation, balancing cost, accuracy, and robustness.

3.2.3 Deep Learning Approaches

As introduced in Section 2.2, deep learning-based object detectors are broadly divided into two-stage architectures (e.g., Faster R-CNN, Mask R-CNN) and one-stage architectures (e.g., YOLO, SSD). In fruit detection research, both families have been extensively evaluated, with the choice between them typically reflecting a trade-off between detection accuracy and inference speed [9, pp. 4-6].

Two-stage detectors have been applied to fruit detection tasks where high precision is critical, especially under heavy occlusion. For instance, Faster R-CNN was used to detect apples in dense-foliage fruiting-wall trees, achieving an average precision of 89% at a processing time of 181 ms per image [9, p. 5]. An attention-augmented Faster R-CNN for coconut cluster detection reached a mAP of 91.2% for unshaded fruits and 88.3% for leaf-occluded fruits [9, p. 6]. Despite their accuracy, the inference speed of two-stage detectors remains a limiting factor for real-time robotic harvesting.

One-stage detectors, particularly the YOLO family, are preferred when fast inference is required. For apple detection, an improved YOLOv5m was designed to distinguish pickable from non-pickable apples, achieving 81% accuracy at 25 ms per image [14, p. 2]. In a comparative study on date fruit detection, YOLOv8n reached a mAP@0.5 of 0.942 with a training time of only 2 hours, while Faster R-CNN required 19 hours to achieve comparable accuracy (mAP@0.5 = 0.94). Moreover, YOLOv8n achieved an inference speed of approximately 50 FPS versus 3 FPS for Faster R-CNN on the same hardware [15, pp. 10-12].

To further improve detection in complex orchard environments, researchers have proposed numerous modifications to the base YOLO architecture. Attention mechanisms such as the Convolutional Block Attention Module (CBAM), which emphasizes informative channels and spatial regions while suppressing irrelevant ones, have been integrated to help the network focus on fruit regions while reducing background clutter [16, pp. 4-5]. Multi-scale feature fusion strategies, including Adaptive Spatial Feature Fusion (ASFF), which learns to weight features from different scales adaptively, have been added to improve detection of small or overlapping fruits [16, pp. 5-6]. Loss function refinements, such as replacing the Complete Intersection over Union (CIoU) loss with the Wise Intersection over Union (WIoU) loss, which introduces a dynamic non-monotonic focusing mechanism to reduce the influence of low-quality samples, have been shown to improve bounding box regression under occlusion [16, pp. 7-8].

Despite these advances, several open challenges remain. Models trained on one orchard often fail to generalize to another due to differences in fruit variety, lighting,

background, and camera perspective. This domain shift problem is exacerbated when only limited real-world training data is available. Furthermore, the need for real-time inference on embedded platforms imposes strict constraints on model size and computational complexity, motivating the development of ever more efficient yet accurate detection architectures. These challenges are addressed in the following sections on localization, synthetic data, and the specific research gaps that motivate the present thesis.

3.3 Fruit Localization and 3D Positioning

Detecting a fruit in a 2D image is only the first step toward autonomous harvesting. To enable a robotic manipulator to approach and grasp a target, the vision system must also provide accurate three-dimensional position information. This section reviews the principal techniques for fruit localization, namely stereo vision, RGB-D cameras, and LiDAR-based methods, with an emphasis on their practical performance in orchard environments.

3.3.1 Stereo Vision Based Localization

Stereo vision systems use two or more calibrated cameras to compute depth through triangulation. By identifying corresponding points in the left and right images, disparity maps are generated and converted to depth maps. This principle was already introduced in Section 2.2. In fruit harvesting applications, stereo vision has been widely adopted because it uses passive sensing and can be implemented with relatively low-cost hardware.

However, stereo matching in orchards is challenging due to texture-poor regions (e.g., uniformly colored fruit surfaces), repetitive patterns (e.g., leaf textures), and occlusions. Under such conditions, correspondence finding often fails, leading to sparse or incorrect depth maps [9, pp. 10-12]. To mitigate these issues, preprocessing techniques such as color-based segmentation and morphological filtering have been applied to improve matching reliability. For example, a stereo system for pineapple fruit achieved

absolute positioning errors of approximately 25 mm at distances of 1.7–2.7 m [9, p. 10]. For citrus localization, a combination of adaptive thresholding and spatial interpolation methods reduced the average positioning error to less than 2.75 mm [9, p. 10].

To increase perception range and robustness, multi-camera stereo setups have been developed. A system using four color cameras with point cloud correction and splicing algorithms improved the accuracy of reconstructed fruit surfaces [9, pp. 10-12]. Despite these advances, stereo vision remains computationally expensive, and its performance degrades under low light or when fruit surfaces are highly specular. Therefore, many modern harvesting robots supplement or replace stereo vision with active depth sensors.

3.3.2 RGB-D Camera Based Localization

RGB-D cameras have become a popular choice for fruit localization because they provide registered color and depth images at video frame rates without requiring complex stereo matching. Two main technologies exist: structured light and time-of-flight. Structured light cameras project a known infrared pattern and analyze its deformation, while time-of-flight cameras measure the round-trip time of emitted light pulses [9, p. 11].

Several studies have demonstrated the effectiveness of RGB-D cameras for fruit localization. A sweet pepper harvesting robot used a time-of-flight camera to extract depth information from detected regions, achieving successful picking [9, p. 11]. For occluded apples, a frustum-based point cloud processing method combined with YOLACT++ reduced the average positioning error by 43% compared to standard depth extraction [9, p. 11]. On kiwifruit, a comparison of different RGB-D cameras showed that a structured-light camera performed best, with a localization speed of 0.062 s per image and an average error within 19 mm [9, p. 11].

RGB-D cameras also enable more sophisticated localization pipelines. For instance, a green apple localization method first analyzed the gradient field of the depth image to find vorticity centers, then used density peak clustering for segmentation, achieving robust detection under challenging illumination [9, p. 11]. For citrus, an approach

combining density clustering and SVM on RGB-D data achieved average localization errors of about 13 mm in all directions [9, p. 11].

Nevertheless, RGB-D cameras have limitations: their effective range is typically 3–5 meters, they are susceptible to strong ambient infrared light (e.g., direct sunlight), and depth accuracy degrades near object edges. For outdoor orchard environments, these factors must be carefully considered.

3.3.3 LiDAR and Point Cloud Approaches

LiDAR sensors offer an alternative that is largely immune to lighting variations. By scanning the environment with laser beams, LiDAR produces dense 3D point clouds that can be used for fruit detection and localization without relying on ambient illumination. This characteristic is particularly advantageous for orchards where sunlight changes rapidly.

A mobile terrestrial laser scanner was used to detect and locate apples, achieving accuracy comparable to RGB-based systems while being unaffected by sunlight changes [9, p. 12]. To guide a cherry harvesting robot, an infrared laser rangefinder was employed to obtain 3D positions of target fruits and obstacles, enabling collision-free trajectory planning [9, p. 12]. A more recent active positioning system combined an RGB camera, an RGB-D camera, and a red line laser on a linear moving stage, achieving depth positioning errors below 4 mm within a range of 0.6–1.2 m, outperforming standalone RGB-D cameras [9, p. 12].

For dynamic environments, a real-time system integrating visual SLAM for camera pose estimation and LiDAR for depth measurement achieved an average positioning error of 21.1 mm [9, p. 12]. This demonstrates that LiDAR can be effectively combined with visual SLAM to maintain accuracy even when the robot or the target is moving.

Despite these successes, LiDAR sensors remain significantly more expensive than RGB or RGB-D cameras, and their data processing is computationally intensive. Consequently, their adoption in cost-sensitive agricultural robots has been limited, although

falling sensor prices and increasing edge-computing capabilities may change this trade-off in the future.

3.3.4 Summary of Localization Approaches

Each localization technology offers distinct trade-offs. Stereo vision is inexpensive and passive but computationally heavy and texture-dependent. RGB-D cameras provide fast, registered depth but suffer from limited range and sunlight sensitivity. LiDAR is robust to lighting and provides dense point clouds but is expensive and computationally demanding. In practice, many successful harvesting systems adopt hybrid approaches, combining an RGB camera for appearance-based detection with an RGB-D or stereo sensor for depth, thereby balancing cost, accuracy, and robustness. The choice of localization method ultimately depends on the specific orchard environment, the required picking precision, and the available computational and financial resources.

3.4 Synthetic Data and Sim-to-Real Transfer

The previous sections have shown that deep learning-based fruit detection achieves high accuracy when abundant, well-annotated real-world training data is available. However, collecting and annotating such data is time-consuming, expensive, and constrained by seasonal harvest windows. Moreover, models trained on one orchard often fail to generalize to others due to differences in fruit varieties, lighting, backgrounds, and camera setups – a problem known as domain shift. One promising solution is to generate synthetic training data using physics-based simulation environments. This section reviews the state of the art in synthetic data generation for fruit detection, the effectiveness of different training regimes, and the remaining challenges for real-world deployment.

3.4.1 Generating Synthetic Orchard Data

Modern simulation platforms enable photorealistic rendering of agricultural scenes with automatic ground-truth annotation [17, p. 2]. By procedurally varying scene param-

eters – including lighting conditions, camera poses, fruit placements, occlusions, and background textures – synthetic datasets can cover a much wider range of visual variations than is feasible with real data collection alone. This technique, known as domain randomization, encourages the trained model to learn domain-invariant features and thereby improves generalization to real-world conditions [17, p. 2].

Two principal strategies for synthetic data generation have emerged. The first emphasizes photorealism: the goal is to make synthetic images visually indistinguishable from real photographs. The second prioritizes structured variation: even if individual images are not perfectly realistic, the model is exposed to such a broad distribution of visual attributes that it cannot overfit to any particular style. In practice, state-of-the-art approaches combine both strategies [17, p. 2-3].

A fully synthetic orchard dataset can be generated using procedural content generation and physically-based rendering. By randomizing tree structure, fruit density, lighting direction, and camera parameters, a single synthetic dataset can contain examples of edge cases – such as backlit fruit, hail damage, or heavy occlusion – that are rare in real-world collections [18, p. 4-6]. Furthermore, synthetic images are automatically annotated with pixel-perfect bounding boxes and, if required, segmentation masks, eliminating human labeling errors.

3.4.2 Training Regimes and Domain Gap

When evaluating synthetic data for fruit detection, three training regimes are typically compared. Real-only training uses exclusively manually annotated real images and serves as an upper-performance baseline under ideal conditions. Synthetic-only training uses only rendered images and tests how well a model can transfer from simulation to reality without any real data. Hybrid training combines both real and synthetic samples, either by joint training from the start or by pretraining on synthetic data followed by fine-tuning on a small set of real images.

A substantial domain gap for synthetic-only models has been observed consistently. For instance, using YOLOv8s on fruit detection, a model trained on 1000 synthetic

images achieved a $mAP@0.5$ of only 0.626 on a real test set, whereas a model trained on only 50 real images reached 0.978 under the same conditions [17, p. 11]. This indicates that synthetic images alone cannot yet replace real data, despite advances in rendering quality.

However, hybrid training strategies significantly close this gap. Adding as few as 50 real images to 1000 synthetic images raised the $mAP@0.5$ to 0.974, nearly matching the real-only baseline [17, p. 11]. Increasing the synthetic dataset size from 1000 to 5000 images brought only marginal additional improvements, suggesting that beyond a certain point, diversity matters more than volume. Interestingly, a pure synthetic approach (without any real data) has been reported to achieve even higher recall than real-only models under relaxed confidence thresholds, at the cost of lower precision [18, p. 7]. This trade-off is particularly relevant for harvesting applications, where missing a fruit (low recall) is more detrimental than a few false positives.

3.4.3 Deployment on Embedded Hardware

An often overlooked aspect of synthetic data evaluation is whether the resulting models can run in real time on the embedded hardware typically used in agricultural robots. Models trained under different regimes can be exported to a standard intermediate representation and optimized for the target hardware using reduced numerical precision (e.g., 16-bit floating point). On a typical embedded GPU platform for edge computing, such an optimized model achieved an inference latency of approximately 19 ms per image, corresponding to around 50 frames per second, and an end-to-end processing time of about 35 ms per image when including preprocessing and postprocessing steps [17, p. 14]. This confirms that models trained with synthetic data can meet the real-time requirements of autonomous harvesting systems without sacrificing detection accuracy.

3.4.4 Open Challenges and Opportunities

Despite these promising results, several challenges remain. First, the effectiveness of synthetic data depends heavily on the quality of the 3D assets and the realism of the rendering pipeline. Second, domain randomization parameters must be carefully tuned; too little randomization fails to close the domain gap, while too much may introduce physically implausible scenes that harm learning. Third, the computational cost of generating large synthetic datasets (e.g., 5000 images) is non-negligible, although it remains far lower than manual annotation of real data.

For the present thesis, these findings motivate a systematic comparison of real-only, synthetic-only, and hybrid training for apple detection using a custom-built simulation environment. In particular, it remains unclear how much synthetic data is needed to match the performance of a real-only baseline when the real dataset is extremely small (e.g., 50 or fewer images), and whether the optimal mixing ratio depends on the specific orchard conditions. These questions are addressed in the experimental part of this work.

3.5 Research Gaps

The preceding review has shown that deep learning-based fruit detection has made substantial progress, yet several important gaps remain.

3.5.1 Scarcity of Datasets

Although several public datasets exist for fruit detection, most are limited in size, annotation quality, or environmental diversity. Many datasets provide only bounding boxes without segmentation masks, and they often lack coverage of challenging conditions such as heavy occlusion, backlighting, or varying fruit maturation stages [12, p. 2-3]. Collecting and annotating real-world orchard data remains expensive, labor-intensive, and constrained by harvest seasons. This data scarcity limits the generalization ability of models trained exclusively on real images [9, pp. 12-13].

3.5.2 Limited Systematic Evaluation of Synthetic Data

Synthetic data generation has been proposed as a scalable alternative to manual annotation. A controlled comparison between real-only, synthetic-only, and hybrid training regimes for apple detection demonstrated the potential of synthetic datasets for improving generalization performance under varying orchard conditions [18, p. 4-5]. Moreover, the optimal ratio of synthetic to real images for apple detection in orchards remains unclear, as does the point at which additional synthetic data yields diminishing returns [17, p. 11]. For the specific case of apple harvesting robots, a quantitative analysis of how synthetic data can reduce the need for real annotations is still lacking.

3.5.3 Domain Shift and Generalization

A recurring finding in the literature is that models trained on one orchard often fail to generalize to another due to differences in lighting, background, fruit variety, and camera perspective [9, p. 13]. This domain shift problem is exacerbated when only a small amount of real training data is available. While hybrid training with synthetic data has shown promise in reducing the domain gap, the extent to which synthetic data can replace real data for cross-orchard generalization has not been fully quantified [17, p. 14-15].

3.5.4 Consideration of Embedded Deployment Constraints

Many fruit detection studies report accuracy metrics on desktop GPUs but do not evaluate whether the trained models can run in real time on the embedded hardware typical of agricultural robots. Real-time inference on edge devices imposes strict constraints on model size, computational complexity, and memory usage [17, p. 4]. Few studies have systematically compared different training regimes (real-only, synthetic-only, hybrid) under the same embedded deployment conditions, leaving a gap between laboratory accuracy and field-ready performance.

4 Overview of Data Sources

4.1 Real-World Data

4.2 Synthetic Data

4.3 Data Annotation

4.4 Data Integration and Harmonization

4.5 Data Splitting Strategy

4.6 Data Augmentation

4.7 Dataset Summary

5 Methodology

5.1 Model Architecture

5.2 Training Pipeline

5.3 Implementation

5.4 Experimental Design



6 Results

6.1 Training Results

6.2 Model Performance

6.3 Qualitative Results

6.4 Comparison of Experiments



7 Discussion



8 Conclusion and Future Work

8.1 Conclusion

8.2 Future Work

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